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DEPARTMENT
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FLY MI

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PILOT OPERATING HANDBOOK (POH)

13 - Fly-Mi - Politecnico di Milano



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1 Introduction

This handbook is intended for operators and pilots of the O.L.I.V.I.A. UAS and serves as a reference for its correct and safe operation within the **X-Plane 11** simulator. It provides essential guidance on operational procedures, system limitations, and manoeuvre constraints, ensuring adherence to best practices in simulated flight conditions. This POH is specifically tailored to the operational characteristics and constraints of the virtual environment. It includes detailed explanations of the aircraft's functionality within **X-Plane 11**, as well as key differences between the simulated model and the final production version. Additionally, the handbook presents critical performance parameters of the UAV derived from numerical simulations and data obtained from the **X-Plane** simulator, offering valuable insights into the expected behaviour of the aircraft.

2 Aircraft Controls

- The aircraft is equipped with conventional flight controls, including ailerons, an elevator, and a rudder. These controls are actuated by electric servomotors, which are integrated due to the aircraft's V-tail configuration.
- The aircraft features flaps with two selectable positions: **Takeoff (TO)** – First-notch position (+1), and **Landing (LND)** – Full flap extension (+2).
- The aircraft is powered by a single electric motor driving a fixed-pitch propeller.
- The aircraft's landing gear (LG) is not equipped with brakes.
- The LG is a fixed tricycle and does not retract.
- The nose LG incorporates a steering mechanism.
- The payload can be released via the release system, which consists of an electric servomotor that retracts a pin, thereby releasing the retention cables securing the payload (PL). Once the PL is released, it is guided along dedicated rails, with its weight causing the cargo doors to open. Meanwhile, the parachute deployment cable remains securely attached to the UAV's structure.
- The **X-Plane** model of the aircraft is NOT capable of performing reverse thrust, which, on the other hand, will be a control only implemented on the real aircraft.

Reverse Thrust: Reverse thrust has been deliberately excluded from the **X-Plane** O.L.I.V.I.A. version, as its inclusion would split the throttle range between positive and negative values. This continuous transition could lead to confusion in control interpretation. The aircraft controls should remain as **intuitive** as possible; therefore, this functionality has been extensively tested in the simulator solely to confirm its feasibility in real flight. Moreover, enabling reverse thrust would require a **dedicated arming system** through the radio control, which would need to be **specifically configured** for this feature. Further details on the system's implementation can be found in Table 6.

3 Pilot Tips

1. The aircraft experiences significant torque effects during take-off. Appropriate right rudder and the LG's nose steering wheel should be used to control the aircraft on the ground.
2. The installed power is considerable. **Full throttle should only be used during the takeoff procedure.** Excessive throttle application during other phases of flight may lead to structural overstress.
3. During cruise, a small amount of throttle should be used to maintain efficient flight and prevent exceeding the never-exceed speed (V_{NE}), as well as to avoid subjecting the aircraft to excessive load factors. More information can be found in the subsection 5.3.
4. The aircraft exhibits high responsiveness in both pitch and roll axes. Control inputs should be smooth and gradual to ensure stable flight.
5. Care should be taken to adhere to flap limitations: deploy flaps only within the approved speed range to prevent potential structural damage.
6. During both take-off and landing, airspeed must be carefully managed to ensure proper rotation and approach manoeuvres, allowing compliance with the designated distances.

4 Limitations

Propulsive System: O.L.I.V.I.A. is powered by a single *T-Motor AT4120 Longshaft KV560* motor. The mounted design propeller is the *APC 15x10E*, although propellers up to a **maximum diameter of 43.2 cm** (17 in) are allowed; the limit is imposed by the landing gear's ground clearance. The following tables outline the critical limitations of the propulsive system.

Motor	Limitations
Max Continuous Power	1800 W (2.414 bhp)
Max Rotational Speed	15,000 RPM
Optimal Range	0 – 10,000 RPM

Table 1: Motor Limitations.

Propeller	Limitations
Rotational Direction	Clockwise (CW)
Max Propeller Speed	12,000 RPM
Optimal Range	0 – 9,000 RPM

Table 2: Propeller Limitations.

Airspeeds: Table 3 presents all airspeed limitations applicable to O.L.I.V.I.A. The various flap configurations referenced in the table are defined as follows:

- *Clean Configuration:* No flap extension, used during cruise and for normal flight operations.
- *TO Configuration:* First-notch (+1) flap extension, employed during takeoff.
- *LND Configuration:* Full (+2) flap extension, employed during landing.

Adherence to these airspeed limits is essential to prevent structural damage and maintain safe handling characteristics. It is important to notice the difference between the terms **MTOM** and **Empty**, which refer to the two main flight conditions: Maximum Take-Off Mass (MTOM) and Empty mass (without payload), respectively.

Symbol	KTAS @MTOM	KTAS @Empty	Speed
V_{NE}	80	80	Never exceed speed
V_{NO}	70	70	Maximum structural speed in rough air
V_{S0}	20	15	LND configuration stall speed
V_{S1}	21	16	TO configuration stall speed
V_S	22	17	Clean configuration stall speed
V_{FE}	35	35	Max speed in LND configuration
V_{FE1}	40	40	Max speed in TO configuration

Table 3: Airspeed Limitations.

Center of Gravity (CG) and Weight: O.L.I.V.I.A. is designed to fly at any weight between the Maximum Take Off Mass, $M_{TOM} = 7$ kg and the Empty Mass (no PL), $M_E = 4.5$ kg.

The UAV's centre of gravity shall remain within the prescribed limits to ensure safe and stable flight characteristics. The CG excursion limits are listed in Table 4. The **aft CG limit** represents the most rearward allowable CG position, corresponding to reduced stability and increased sensitivity to control inputs. The **forward CG limit** is restricted to ensure sufficient elevator authority for pitch control and ground manoeuvrability. The **optimal CG position** is recommended for achieving a balanced flight performance, providing adequate stability without compromising control responsiveness. The **Static Margin (SM)** mentioned in Table 4 is a measure of longitudinal stability that can be calculated with the following formula:

$$SM = \frac{X_N - X_{CG}}{MAC}$$

where X_N is the neutral aerodynamic point of the aircraft, estimated at 0.145 m from the wing's leading edge (LE) in the root section. X_{CG} is the center of gravity position.

CG Position	From Wing LE [%MAC/mm]	From Motor Mount [mm]	Static Margin [%]
Forward Limit	9.1/25	415	27
Optimal	45.5/125	515	16
Aft Limit	50.4/140	530	10

Table 4: Center of Gravity Limits.

Manoeuvring: The V - n diagram illustrates the limitations of manoeuvring flight by showing the relationship between *load factor* and *airspeed*. This diagram highlights:

- **Stall Speed Limitation:** Defined by the lower boundary of the curve, indicating where the aircraft will stall at different load factors.
- **Structural Limits:** The aircraft shall not exceed the maximum positive load factor of $n_{\max} = 2.2$ during normal operations.
- **Ultimate Load Factor:** The absolute structural limit is $n_{\text{ultimate}} = 3.0$, beyond which structural failure may occur.

The pilot is responsible for operating within these limits and maintaining a safe margin from excessive load factors and speeds.

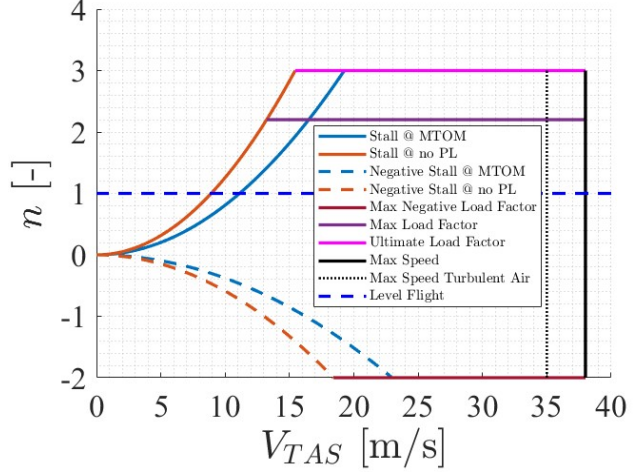


Figure 1: V vs n Diagram (Clean Configuration).

Operational Restrictions: The following operational limitations are outlined to ensure the safe operation of the aircraft. Failure to comply may result in compromised flight safety and reduced aircraft performance.

- Flights are not allowed in heavy rainfall or when wind conditions exceed 20 knots, with gusts reaching up to 25 knots. Take-off and landing are also prohibited if the runway's crosswind component exceeds 5 knots, with gusts up to 8 knots.
- The aircraft shall not be operated if the outside air temperature (OAT) exceeds 40°C.
- Take-off and landing with negative (-) flap settings are prohibited.

5 Operating Procedures

5.1 Pre-Flight Checks

1. Ardupilot parameter list + Mission waypoints
2. Shake the wing to avoid pinching servo cables with the bayonet
3. Check the fuselage tube connector while inserting in wing-box
4. Insert carefully wing connectors and check for pin retraction
5. Alcool on ESC
6. Clean Lidar surface
7. **Check** servo connector correctly inserted
8. Tail connector fastened
9. Condition check
 - (a) Visual inspection of the plane (intact, clean and working components):
 - i. Landing gear (LG)
 - ii. Structural integrity
 - iii. Fuselage (no loose components)
 - iv. Payload release correctly retained
 - v. Clean aerodynamic surfaces
 - vi. No holes in Oracover
 - (b) Rear capote sensors and antennas integrity check
 - (c) Front capote electronics integrity check
 - (d) Securely tightened bolts, retention mechanisms and propeller
 - (e) Check connectors condition and electronic components correctly fastened
 - (f) Wing retention pin inserted
 - (g) Correct movement of all moving parts:
 - i. Servomotors
 - ii. Control surfaces
 - iii. Payload release system
 - iv. Nose landing gear
10. Batteries check (SOC and correct retention):
 - (a) Radio battery
 - (b) Ground Control Station (GCS) / PC battery
 - (c) Main battery (main switch off)
11. CG check through balances for correct stability margin (see Table 4)

5.2 Startup Procedures

1. Main switch connected
2. **SD card inserted**
3. GCS connected
4. GCS correctly functioning
5. Sensor calibration:
 - (a) IMUs completed
 - (b) Compass completed
 - (c) GPS completed
 - (d) Pitot completed
6. Check for NO errors from Ardupilot
7. Correct GPS signal
8. Correct LIDAR reading
9. FC core temperature check
10. ESC temperature check
11. SOC FC battery
12. SOC main battery from Radio
13. Correct autopilot output direction on control surfaces for safe flight (check if outputs are inverted)
14. Mission planner mission upload completed
15. Check written waypoints **one-by-one !**
16. Check written parameters
17. Correct first waypoint selected
18. Failsafe test completed
19. Temperature check ESC + FC
20. Pitot cover and other red-tagged items removed
21. Engine properly working

Take off Procedure: *Flaps* shall be set to the *first-notch (+1) position*. The take-off manoeuvre shall be performed by applying *full throttle* (80 A of current) while simultaneously applying *right rudder* to counteract the torque generated by the engine. The control stick shall only be pulled back after reaching V_{S1} (IAS) and preferably before exceeding the take-off box distance limitation of 10 m.

Cruise Procedure: The pilot shall stabilize the aircraft at either the best-range airspeed or the minimum sink rate speed as explained in Table 5. Trim of the elevator might be necessary depending on the loading condition of the UAV.

Payload Release Procedure: The payload shall be released as close as possible to the designated GPS coordinates. The pilot shall remain vigilant during this phase for potential release failures. Slight pitch variations might occur performing the manoeuvre, but should be well controllable, as the payload's centre of gravity is designed to be nearly coincident with the aircraft's structural centre of gravity.

Landing Procedure: The pilot shall reduce the airspeed below V_{FE} , extend *full flaps* (+2), and align the aircraft with the runway. The throttle shall be used solely to stabilize the descent. The pilot should aim to touch down at the beginning of the designated landing box.

5.3 Flight Performance

Recommended Airspeeds: The optimal airspeeds for the primary flight procedures are outlined in Table 5. However, it is important to note that the theoretical best range and minimum sink-rate speeds are, in reality, relatively close to the stalling speed. Therefore, to maintain a safe operational margin, an additional 2.5 m/s has been factored in. The battery current draw at the best-range attitude is expected to correspond to approximately 7 A (MTOM) and 5 A (Empty), depending on battery charge and atmospheric conditions.

Flight Condition	MTOM [kn]	Empty [kn]
Best Range	30.1	25.2
Minimum Sink Rate	27.6	23.1
Flap (+1) TO rotation	22.2	16.9
Flap (+2) LND approach	22.0	16.5

Table 5: Recommended Cruise Airspeeds.

TO and LND Performance: O.L.I.V.I.A. is designed for taking off and landing on grass runways. The TO is designed to be performed in less than 10 metres and the landing, touching down and coming to a full stop, in less than 20 metres. The effect of wind direction and intensity present on the runway *should be taken into account* when performing these manoeuvres, calculating and considering them carefully beforehand.

Loading Capability: O.L.I.V.I.A. offers various possible payload loading configurations. Figure 2 illustrates how the total range varies as a function of the on-board payload. The calculation assumes flight at the best range speeds as indicated in Table 5, that the payload is dropped exactly at half the total covered distance and that the battery is only usable up to a state of charge (SOC) of 20%, to ensure sufficient safety margin. The minimum payload is the *ADB*, which has a mass of 0.5 kg. For extra safety, the total energy estimation also takes into account a *go-around* loop, which will be necessary in case the first landing attempt is not successful. The orange dashed line represents the estimated total distance travelled in the competition scenario.

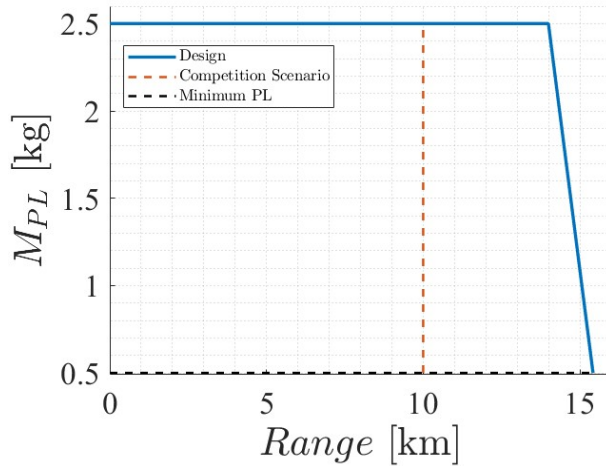


Figure 2: Payload - Range Diagram.

6 Design Modifications and X-Plane Adjustments

The table below summarises all the differences between the **X-Plane** model and the actual O.L.I.V.I.A. design. The *Notes* column outlines the reasons for these differences, as well as how they will be implemented in the real aircraft.

X-Plane Limitation	Notes
Aerodynamics	Approximation of the polar curve and pitching moment using <i>blade element theory</i> . The theory slightly underestimates the aerodynamic drag of the lifting surfaces.
Surface Planform	The software does not allow an exact reproduction of a semi-elliptical wing planform or precise tail planform due to the limited number of wing stations.
Surface Thickness	Wing and propeller thickness are not reproduced exactly since Airfoil Maker allows only 14-node airfoils.
Propeller Model	The propeller airfoils are interpolated from root to tip in a similar, though not exact, manner due to limitations in the simulation model.
Propulsion System	The engine and propeller coupling is fine-tuned as accurately as possible, but due to the inability to directly adjust the motor's KV rating, some errors might remain.
Motor mounting angle	The motor is currently mounted onto the fuselage without any angle. Further studies and simulator testing will determine if a specific mounting angle is necessary to improve handling characteristics.
High RPM	Slightly higher RPM than expected due to limitations in the motor model. However, all other propulsion parameters (thrust, power, current, and voltage) are consistent with test data.
Battery	The battery is tuned based on real discharge tests. However, the internal battery physics cannot be accurately modelled, resulting in a slightly shorter battery life compared to real-world performance.
Thrust Reverse	Thrust reverse will be implemented on the real O.L.I.V.I.A. model via a switch on the radio control, using ArduPilot for safety reasons. This functionality has been extensively tested in X-Plane to verify its feasibility and to ensure the aircraft can land within 20 metres.
Landing Gear	The landing gear structure section is modelled approximately, as it would require a complex "C-shaped" section. Additionally, the type of tyres has not yet been chosen, so future modifications should be expected.
Sensors	The Pitot tube and the Lidar sensor have not been modelled in X-Plane . However, their measurements have been used for model tuning and simulated using the radio altimeter and airspeed gauge in the cockpit.
PL release system	The PL is simulated as water in X-Plane . The PL release system is not modelled, and therefore, neither is the relative cargo door opening mechanism.
Wing flexibility	The wing flexibility has been estimated performing some finite element simulations (FEM) on the full plane structure. Although this analysis has been validated and verified, the results might differ slightly from reality.

Table 6: X-Plane vs Real model differences.